



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000
WEEK 4.01

TUTOR GUIDE

The cases and materials in this guide were written and developed by;

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The cases and materials in this guide were reviewed by;

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RELEVANT PRIOR LEARNING

MED1000

1.21 Applied anatomy of the Airway (L)

MED2000

2.01 Adult Intubation (CCS)

2.22 Paediatric Intubation (CCS)

2.29 Rural Trauma Week (Workshops & lectures)

MED3000

3.02 Cases in Acute Dyspnoea (CCS)

3.21 Paediatric Intubation (CCS)

WEEK 01_ ANAESTHESIA		
Yr	Wk	LEARNING OBJECTIVES
4	1	• Know the key components of history and examination when assessing a patient for whom a general anaesthetic is planned for an elective procedure.[ie low risk] • Understand the importance of assessing exercise tolerance in patients prior to a general anaesthetic. • Know how to assess exercise tolerance in a patient for whom a general anaesthetic is planned. • Outline the investigations that are required as part of the assessment for a patient for whom a general anaesthetic is planned for an elective procedure and be able to explain the rationale for these investigations. • Interpret the results of investigations in the context of patient requiring a general anaesthetic and be able to appropriately modify an anaesthetic plan as a result of investigation results.
4	1	• Know the key components of the assessment of a patient for general anaesthetic with a recent history of severe illness. [ie high risk] • Outline the investigations that are required as part of the assessment of a high risk patient for whom a general anaesthetic and be able to explain the rationale for these investigations. • Define general anaesthesia. • Detail the components of 'balanced general anaesthesia' • Explain the term minimum alveolar concentration [MAC] in relation to inhalational anaesthetic agents. • Outline the factors that can increase and decrease MAC. • Explain the mechanism of action, pharmacokinetics, possible adverse reaction and drug interactions of the following inhalational anaesthetics – Isoflurane, sevoflurane, desflurane and nitrous oxide. • Explain the mechanism of action, pharmacokinetics, possible adverse reaction and drug interactions of the following intravenous anaesthetic agents' propofol, thiopentone and ketamine & midazolam. • Explain the mechanism of action, pharmacokinetics, possible adverse reaction and drug interactions of the following opioids, morphine, fentanyl and Alfentanil and non opioid tramadol. • Explain the mechanism of action, pharmacokinetics, possible adverse reaction and drug interactions of the following muscle relaxants, suxamethonium, Rocuronium, Vecuronium, cisatracurium.

WEEK 01_ANAESTHESIA		
Yr	Wk	LEARNING OBJECTIVES
4	1	• Know the key components of the assessment of a patient for general anaesthetic with a history of respiratory illness eg asthma. • Outline the peri-operative management plan of patients with reactive airways disease who require general anaesthesia. • Outline the factors that need be considered when obese patients require general anaesthesia. • Know the anatomy of the airway from the oral and nasal cavities to the trachea, including the sensory innervations of upper and lower airway. • Outline the key steps in preparation for airway management prior to intubation for general anaesthesia in an elective procedure. • Know the key equipment that is used in airway management and be able to identify and briefly outline the use and function of the following equipment endotracheal tube, oral airway, nasopharyngeal airway, laryngeal mask, bougie, stylet and magill forceps. • Know the indications for endotracheal intubation. • Know the potential complications of endotracheal intubation. • Outline the process for confirming correct placement of the endotracheal tube.
4	1	• Know the key components of the history and examination in the assessment of a trauma patient for an emergency general anaesthetic. • Outline the key features that contribute to aspiration risk in a patient undergoing an emergency general anaesthetic. • Outline the key features that contribute to potentially difficult airway management in a patient undergoing an emergency general anaesthetic and how to optimise patient preparation. • Describe the principles of a rapid sequence induction for an emergency general anaesthetic.

CASE ONE

Short case number: 4_01_01

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Pre-operative assessment clinic.

Topic: Pre-operative Anaesthesia assessment_low risk

CASE



Harry is a 53 year old man who presents to the Anaesthetic Clinic for assessment prior to his elective left knee arthroscopy. His knee is painful, particularly at the end of the working day, and is intermittently swollen and hot.

Booking observations are: T 36.6; BP 120/80; PR 84reg; RR 16; SpO2 97% in RA

His medical history includes treated hypertension and gout. He is an ex-smoker, ceasing 10 years ago, and has an average of 5 standard alcoholic drinks per week.

His medications are amlodipine 10mg mane and allopurinol 300mg mane.

QUESTIONS

1. What are the key components of a pre-anaesthetic history and physical examination?
2. Why is exercise tolerance important? How can it be assessed?
3. What investigations are relevant in this case and why?
4. What factors increase "aspiration risk"?
5. What is the impact of cigarette smoking on peri operative care and anaesthesia?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.
- Duke 2000 Anaesthesia Secrets Second Edition

Electronic resources: [also available on blackboard]

ANZCA Professional Documents: <http://www.anzca.edu.au/resources/professional-documents/>

- PS7 Recommendations on the Pre-Anaesthesia Consultation
- PS12 Statement on Smoking as Related to the Perioperative Period
- PS15 Recommendations for the Perioperative Care of Patients selected for Day Case Surgery

ANSWERS

1. What are the key components of a pre-anaesthetic history and physical examination?

The goal of pre-anaesthetic assessment is to **formulate an anaesthetic plan**, establish rapport with the patient and obtain informed consent. History, physical examination and review of investigations helps identify for each individual case the patient, anaesthetic and surgical factors that will influence the selection of optimum anaesthetic technique and peri-operative care.

To achieve this, one must know:

- (i) the state of the patient presenting
ie. features of the presenting complaint and background co-morbidities
- (ii) the surgical requirements for the case
eg. muscle relaxation, PPV, prone positioning, emetogenic procedure etc.
- (iii) the relevant anaesthetic issues
eg. **known or anticipated difficult airway** management, **history of post-operative nausea and vomiting (PONV)**, **history of malignant hyperthermia etc.**

These components form the basis of a comprehensive pre-anaesthetic assessment, as can be seen in most pre-printed anaesthetic assessment forms.* include form?

Let's look in detail at Harry's case:

HISTORY

Presenting complaint: non-specific knee pain for investigation, likely osteoarthritic in nature
No evidence of generalized sepsis on observations.

Past Medical History: treated hypertension with acceptable booking BP, history of episodic gout
Further information can be gained by enquiring specifically about adequacy of BP control and potential end-organ consequences. For example, "Is your GP happy with your blood pressure?" "Has your dose of antihypertensive been changed recently?" "Have you ever had any complications related to BP control, such as problems with your heart, kidneys or a stroke?"

Systems review should enquire about key conditions in the affecting the major organs/systems. Questions should be broad initially:

"Have you ever had any problems with your **heart, a heart attack or angina?**"

"Do you have any chest or **lung complaints, asthma or emphysema?**"

"Have you ever had any, strokes, fits or turns?"

"Do you have diabetes?"

"Do you suffer from indigestion, heartburn or gastro-oesophageal reflux?"

"Have you ever had any problems with your liver or kidneys in the past?"

Further detailed questioning will follow as indicated.

2. Why is exercise tolerance important? How can it be assessed?

Assessment of **exercise tolerance** gives clues to the patient's cardiorespiratory reserve. It is assessed by convention using a **metabolic equivalent scale** (see table below). **Patients unable to meet the demand of four metabolic equivalents (4METs = climbing a flight of stairs) are considered to have minimal cardiorespiratory reserve and at high risk of perioperative cardiorespiratory compromise.** Sometimes patient's exercise tolerance is limited by other conditions, such as back pain or claudication, such that history alone is inadequate to assess cardiac reserve. These patients may require further non-invasive or invasive cardiac testing such as stress echocardiography or coronary angiography.

Table 3. Estimated Energy Requirements for Various Activities

1 MET	Can you ... Take care of yourself?	4 METs	Can you ... Climb a flight of stairs or walk up a hill?
	Eat, dress, or use the toilet?		Walk on level ground at 4 mph (6.4 kph)?
	Walk indoors around the house?		Run a short distance?
	Walk a block or 2 on level ground at 2 to 3 mph (3.2 to 4.8 kph)?		Do heavy work around the house like scrubbing floors or lifting or moving heavy furniture?
4 METs	Do light work around the house like dusting or washing dishes?		Participate in moderate recreational activities like golf, bowling, dancing, doubles tennis, or throwing a baseball or football?
		Greater than 10 METs	Participate in strenuous sports like swimming, singles tennis, football, basketball, or skiing?

kph indicates kilometers per hour; MET, metabolic equivalent; and mph, miles per hour.

*Modified from Hlatky et al,¹¹ copyright 1989, with permission from Elsevier, and adapted from Fletcher et al.¹²

Exercise tolerance also gives clues to the degree to which the patient's presenting complaint is impacting on their life and their ability to care for themselves post-operatively. These factors are important for assessing suitability for ambulatory (or day-case) surgery, discharge planning and the ultimate suitability of some procedures for individual patients.

Move on to questioning about smoking and alcohol intake, allergies/adverse drug reactions and current medications including non-prescription or complementary therapies.
Knowledge of fasting status is essential.

Past Surgical History: may reveal problems with previous anaesthetics. It is critical to ascertain a history of issues such as allergic reactions, airway difficulties, awareness under anaesthesia or malignant hyperthermia. A positive **family history** for "problems with anaesthetics" should be explored as this may represent malignant hyperthermia or delayed recovery from suxamethonium.

EXAMINATION

Airway examination is commonly the first step. A number of key anatomical assessments are made to help gauge the ease or difficulty anticipated in managing the patient's airway.

These include:

- (i) **general assessment: body habitus, resp distress, on oxygen therapy, stridor, scarring or pathology of the head and neck**
- (ii) mouth opening and interdental distance
- (iii) state of dentition: poor dental health, loose teeth, protruding, prostheses or dentures
- (iv) Mallampatti score: appearance of the posterior pharynx with mouth open and tongue protruding can predict difficulty visualising the larynx at laryngoscopy
- (v) mandibular size and TMJ function
- (vi) thyromandibular distance
- (vii) cervical spine range of motion

Mallampati classification



Further physical examination should be focussed on the cardiac and respiratory systems, with other systems emphasized as indicated by the patient or the case. **For example, in a patient having a carotid endarterectomy it would be prudent to document any pre-existing neurological deficits prior to anaesthesia.**

INVESTIGATIONS

Ordering investigations should be guided by the features of both the patient and the proposed surgery.

General principles include:

- (i) Baseline full blood count(**FBC**), **urea and electrolytes(U and E's)** for older patients having moderately invasive procedures, particularly on medication for cardiovascular disease (eg diuretics that can cause electrolyte disturbances)
- (ii) **Coagulation profile** and group and hold if bleeding diathesis (innate or therapeutic) or procedure associated with potential significant blood loss
- (iii) Liver function tests if history suggestive of hepatic dysfunction
- (iv) Pregnancy test if possibility of pregnancy, particularly with gynaecological surgery

- (v) ECG routine for males >50years, females > 60years, and for any patient with history of cardiovascular disease or arrhythmia
- (vi) CXR if history of cardiovascular or respiratory disease, or malignancy
- (vii) Respiratory function tests (including peak flow measurement, spirometry pre- and post-bronchodilation and lung volume/diffusion capacity evaluation) if significant respiratory disease, particularly recent acute exacerbation
- (viii) Non-invasive and Invasive cardiac investigations: see question 2.

Therefore, in Harry's case it would be reasonable to request FBC and U and E's, and an ECG. One could argue as an ex-smoker with treated hypertension that a CXR may be warranted, but in the absence of any history suggestive of respiratory compromise or cardiac failure, it would **be unnecessary**.

4. **Aspiration** is the passage of liquid or particulate matter from the upper airway into the trachea and lower respiratory tract. **This can result physical obstruction of large and small airways as well as chemical pneumonitis**. Both conditions impair ventilation and gas exchange, and can lead to pneumonia. Risk of this occurring increases with:

- (i) impaired airway protective reflexes
- (ii) depressed level of consciousness
- (iii) vomiting or regurgitation
- (iv) trauma/bleeding in the airway
- (v) delayed gastric emptying: gastroparesis, recent opioid administration
- (vi) incompetent lower oesophageal sphincter(LOS): GORD, hiatus hernia
- (vii) increased barrier pressure at the LOS: obesity, pregnancy, ascites
- (viii) recent ingestion of food

Aspiration risk should be assessed for each patient. Appropriate fasting times for elective surgery are:

Solid food, including milk and infant formula – 6 hours

Breastmilk – 4 hours

Clear liquids – 2 hours

Regular medications are taken with a small sip of water as advised by the anaesthetist. Fasting patients should abstain from chewing gum or suck lollies as this can stimulate increased gastrointestinal secretions.

Aspiration risk can be minimized in elective surgery by pre-medication with appropriate duration of fasting, motility agents to enhance gastric emptying, proton pump inhibitors and antacids to raise gastric pH, attention to patient position and airway management.

5. Cigarette smoking results in:

- (i) **increased blood concentration of carbon monoxide (CO) which reduces the oxygen carrying capacity of the blood**
- (ii) **blood nicotine concentration increases BP, myocardial contractility and HR and causes peripheral vasoconstriction**
- (iii) **hypersecretion of airway mucus, impaired tracheobronchial clearance and airway irritability**
- (iv) Smokers' polycythaemia, increased blood viscosity and hypercoagulable state

These changes contribute to a higher risk in smokers of respiratory complications and poor wound healing.

Blood levels of CO and nicotine improve within 12-24 hours of abstinence. **Sputum production will initially increase for 1-2 weeks following smoking cessation, but declines within 4 weeks**. Hypercoagulable state can take 2 -14 days to reverse potentially increasing the risk of venous thrombosis.

Post-operative respiratory complications have not been shown to decrease until 6-8 weeks. Current recommendations suggest patients should be encouraged to cease smoking 6-8 weeks prior to surgery, or abstain from smoking for at least 12 hours before operation.

CASE TWO

Short case number: 4_01_02

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Pre-operative assessment clinic

Topic: Pre-operative Patient Assessment_High Risk

CASE

You are asked by your consultant to arrange preoperative workup for a 75 yo man who is booked for open Abdominal Aortic Aneurysm resection and graft. The consultant wants to get the surgery done as soon as possible, as the aneurysm measures 8cm in diameter and is a little tender to palpation – he is concerned about imminent rupture.

The patient is an ex-smoker of 30 pack-years; he has hypertension treated with Ramipril and Metoprolol; Type 2 diabetes for 14 years, currently treated with insulin, protaphane 22 units in the morning and 12 units at dinner time; he says he is quite active but on questioning, his only real activity is a 1km round-trip level walk to the TAB 4 days a week, after which he feels quite tired and normally has a sleep; he denies any symptoms suggestive of ischaemic heart disease. His blood tests done yesterday by the surgeon show normal FBC; K of 5.5, Urea 10.0 mmol/l and Creatinine 140 micromol/l.

QUESTIONS

1. Over and above those issues pertinent to all patients, what are the particular preoperative issues for this man?
2. Discuss a plan for preoperative cardiac evaluation, given that this is a patient undergoing high-risk surgery who has limited functional reserve.
3. What is the current role of preoperative beta-blockade in a patient like this?
4. Discuss some possible options for his perioperative diabetes management.

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.

Electronic resources [also available on Blackboard]

- Fleisher LA et al ACC/AHA 2007 Guidelines on Perioperative Cardiovascular Evaluation and Care for Noncardiac Surgery: Executive Summary Circulation 2007 116(17)

<http://circ.ahajournals.org/cgi/content/full/116/17/1971>

ANSWERS

1. High risk surgery (major vascular. Other high-risk includes thoracic, upper abdominal). High-risk patient (multiple co-morbidities, poor functional reserve). He surgery is urgent (as distinct from emergent) as a tender 8cm AAA is a condition at great risk of rupture and sudden death. There is time for preop evaluation of cardiac risk but not over weeks, more like a day or 2, and the testing should be arranged as an inpatient. He is at high risk of having coronary artery disease. His Renal function is abnormal and there is considerable risk to it during and after the proposed surgery. Diabetes needs special consideration.. He has a long smoking history and his respiratory function will be an important determinant of post-op recovery after this major upper abdominal surgery. A discussion of analgesia options is important at the preoperative assessment by the anaesthetist, perhaps not so much by the surgical team members.
2. A slow walk 1km after which he is exhausted suggests 4 METS of activity – quite limited. Unless emergency surgery, he should have more extensive workup even in the absence of other symptoms. DM2 is associated with “silent” myocardial ischaemia and RBBB and Beta-blockade make resting and stress ECG uninterpretable. A few options – aim is to assess risk of perioperative adverse cardiac event (AMI, CCF, death) and to monitor appropriately. To do this, assess coronary circulation, and myocardial function – preferably myocardial function while myocardium is being stressed. If good functional reserve, no symptoms, normal resting ECG, reasonable to proceed. Otherwise, either Dipyridamole-Sestamibi Scan or Dobutamine Stress Echo. DSE gives more assessment of cardiac function; the jury seems to be out. Coronary angio as a **screening** tool risks outweigh benefit.
3. A number of small studies and one larger (POISE) over the last 10 years or so of periop beta blockers and high-risk patients. Inconsistent results for various reasons, largely methodological I think. This man is already on Metoprolol, well beta-blocked with HR 52/min. He should remain on it up to and including the morning of surgery.
4. His Ramipril should be withheld – certainly on the morning of surgery and probably from now, at least until renal artery stenosis can be excluded. It may be contributing to his impaired renal function. Advice from a nephrologist should be obtained, as well as to warn the team of the possibility of post-p renal failure. Possible causes of his renal impairment include, vascular disease, diabetes, hypertension and others. Meticulous attention to haemodynamics intra and post operatively, avoidance of nephrotoxins, eg iv contrast, aminoglycosides, NSAIDs, avoidance of preop dehydration by giving IV fluids while fasting are important.
5. Various options for DM. Aims are prevent hypoglycaemia, and prevent complications of hyperglycaemia. In DM1, prevent ketoacidosis. Not a risk in DM2. Conflicting evidence of benefits of “tight”(4-6) vs “conventional”(5-10) glucose control. For DM2 simplest is no insulin morning of surgery, also no oral hypoglycaemics. Check BGL intraop and if necessary run infusion of soluble insulin to keep BGL < 8 or so. If insulin is given , also need iv of 5% Dextrose available. For DM1,

need some insulin otherwise get ketoacidosis. Either $\frac{1}{2}$ normal am dose, given as soluble insulin, along with 5% D infusion, or run Glucose/Insulin +/-Potassium infusion – there are institutional protocols. If in doubt call patient's endocrinologist. Always stop Metformin 1-2 days preop especially with fasting.

CASE THREE

Short case number: 4_01_03

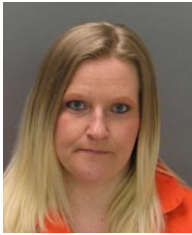
Category: Clinical Skills

Discipline: Anaesthesia

Setting: Pre-operative assessment clinic

Topic: Balanced General anaesthesia.

CASE



Lisa Graham, aged 36 years, is booked for a laparoscopic cholecystectomy. She has documented cholelithiasis, and was hospitalized 6 weeks ago with acute cholecystitis and pancreatitis.

Lisa has been told she will need a general anaesthetic and is concerned about what this entails. She experienced severe nausea, vomiting and muscle pain following emergency surgery for a ruptured appendix 2 years ago.

QUESTIONS

1. What is general anaesthesia?
2. What are the components of “balanced general anaesthesia”
3. Outline the pharmacology of the common agents used to provide balanced general anaesthesia.
4. Provide Lisa with a likely explanation for the experience she had with her appendicectomy?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.

Other resources:

- Smith and Sasada 1999 Drugs in Anaesthesia and Intensive Care Oxford Medical Publications

ANSWERS

What is general anaesthesia?

General anaesthesia is a temporarily induced state of depressed consciousness designed to facilitate surgery or other painful procedures.

What are the components of “balanced general anaesthesia”

1. Balanced anaesthesia refers to the technique of combining drugs from the hypnotic, analgesic and muscle relaxant classes to provide the optimal conditions (of depressed conscious state, amnesia, pain relief and muscle relaxation) required for the proposed surgery or intervention in a given individual patient.

In order to choose this combination, the anaesthetist will take into account:

Patient factors ie. High aspiration risk requiring secured airway with ETT

Surgical factors: ie. Requirement for muscle relaxation and IPPV to facilitate pneumoperitoneum during laparoscopic cases.,

Anaesthetic factors ie. Duration of drug action, mode of metabolism etc

Outline the pharmacology of the common agents used to provide balanced general anaesthesia

Intelligent use of drugs providing the elements for balanced anaesthesia requires an understanding of mechanism of action, dose, pharmacokinetics, adverse reactions/toxicity and drug interactions.

- Inhalational anaesthetics

- Intravenous anaesthetics

- Analgesics

- Neuromuscular junction Blockers

HYPNOTICS

INHALATIONAL ANAESTHETICS

- produce generalised suppression of consciousness historically known as hypnosis
- mechanism of action incompletely understood
- advantages of:
 - gradual induction of anaesthesia via inhaled route with maintenance of spontaneous respiration
 - reliable maintenance of anaesthesia via inhaled route
 - minimal metabolism in the body
- disadvantages of:
 - pungent, a/w irritant
 - malignant hyperthermia triggers

MAC - minimum alveolar concentration

- the concentration of agent in 100% oxygen at 1 atm that abolishes motor response to a standardised noxious stimulus in 50% of patients
- comparative measure of potency for inhalational agents
- multiple factors influence MAC

Increase MAC Children Hyperthermia Hypernatramia Chronic alcoholism	Decrease MAC Extremes of age Hypothermia Hyponatraemia Opioids / Other drugs
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MAC awake = 0.33MAC

- 50% of patients will regain consciousness/follow simple commands

INHALATION AGENT	MAC (%)	FEATURES
Isoflurane	1.15	Bronchodilator
Sevoflurane	1.7	pleasant smell, non-a/w irritant
Desflurane	6	rapid onset/offset, a/w irritant, SNS stimulant
Nitrous oxide	104	potent analgesic, implicated in PONV, dilutional hypoxia

INTRAVENOUS ANAESTHETICS

- produce generalized suppression of consciousness
- convenience of rapid onset via intravenous route
- advantages of:
 - rapid onset and offset of anaesthesia
 - can be adapted for maintenance of anaesthesia

	PROPOFOL	THIOPENTONE	KETAMINE
Mechanism of Action	GABA mediated inhibition of CNS	GABA mediated inhibition of CNS	NMDA R antagonist
IV Induction Dose	2-3mg/kg	3-5mg/kg	1-2mg/kg
Features	Pain on injection Excitatory Phenomena	Venous Irritant PPTE Porphyria CVS stability	Potent Analgesic Dissociative Anaesthetic Hallucinations

BENZODIAZEPINES

- sedative, anxiolytic, amnestic and antiepileptic via enhancement of GABA transmission in the CNS
- negative attributes of this class, such as sleep disruption and addiction potential, should confine their use to specific circumstances only

Midazolam

- water soluble, short-acting BZD
- sedative dose 0.05-0.1mg/kg
- t_{1/2} 1-4hrs
- active metabolites

ANALGESICS

OPIOIDS

- agonists at endogenous mu opioid receptors to provoke analgesia
- other mu opioid receptor mediated responses include: sedation, respiratory and cough reflex suppression, miosis, pruritis, nausea and vomiting, decreased GI motility, urinary retention, tolerance and dependence
- potential side effects via non-opioid receptor actions: central SNS suppression, histamine release, allergy
- numerous formulations for use via numerous routes ie. intravenous, neuraxial, subcutaneous, oral, transdermal and more....

DRUG	MORPHINE	FENTANYL	ALFENTANIL
Features	Potent Analgesia	Potent Analgesia	Potent Analgesia
Dose	0.1 – 0.15 mg/kg	2-4mcg/kg induction dose	10-20mcg/kg ind'n dose
Half Life	2-4hrs adults	3-4hrs	1-1.5hrs
Onset	10-15 min	2-5min	90sec
Duration	3-4hrs	30-60min	5-10min
Metabolites	Active and toxic metabolites	No active metabolites	Active metabolites

NON-OPIOIDS

Tramadol

- multimodal moderate potency analgesic agent
- combined action 1/3 mu opioid receptor agonist, 2/3 monoamine (5HT and NA) uptake inhibition
- oral and IV formulation ****100mg IV approximates the analgesic action of 10mg morphine parenterally
- dose 50 -100mg q6h maximum 400mg per 24hrs
- active metabolite
- side effects of nausea and vomiting, sedation, dysphoria, lowered seizure threshold, dizziness and postural hypotension
- drug interactions: TCA's, SSRI's, MAOI's, CBZ, ondansetron

NSAIDS

- wide range of chemical classes
- unifying mechanism of action is cyclo-oxygenase inhibition, thereby preventing the genesis of inflammatory mediators (eg prostaglandins)
- group adverse reactions include: gastric erosion, nephrotoxicity, platelet function inhibition, bronchospasm, headache, confusion, sedation, hepatotoxicity, tocolysis, closure of PDA and hypersensitivity reactions

PARACETAMOL

- simple analgesic and antipyretic
- safe and well tolerated
- dose adults 10-30mg/kg (1gm) q6h maximum 4gm per 24hrs
children 10-20mg/kg q4h maximum 90mg/kg per 24hrs
- adverse reactions include: GI upset, hypersensitivity reactions, hepatotoxicity in overdose

MUSCLE RELAXANTS/NEUROMUSCULAR JUNCTION BLOCKERS

- competitive antagonists at the NMJ that temporarily disable muscle contraction
- depolarising and non-depolarising types - describes action at muscle end-plate
- will render patient unable to spontaneously breathe or protect airway
- must not be administered to a conscious patient

Suxamethonium

- the only depolarising NMJB
- stored at 4 degrees to maintain activity
- intubating dose 1-2 mg/kg
- rapid onset 45 - 60secs
- short duration of action 5-10min
- these properties make it ideal for use in patients where rapid airway control is required (ie. aspiration risk) or rapid recovery of spontaneous respiration is required (ie. difficult airway scenario)
- adverse reactions:
 - hyperkalaemia
 - post-op myalgia
 - raised intracranial and intraocular pressure
 - malignant arrhythmias
 - prolonged apnoea
 - anaphylaxis
 - bradycardia
 - malignant hyperthermia
 - masseter spasm
- contraindication/precautions:
 - malignant hyperthermia history
 - burns/denervation injury/myopathies/muscular dystrophies
 - plasma cholinesterase deficiency/sux apnoea
 - renal impairment/hyperK+

Non-Depolarizing NMJB's

- competitive antagonists at the NMJ
- short, intermediate and long acting agents with various properties

	Rocuronium	Vecuronium	Cisatracurium
Dose	Intubating Dose - .6mh/kg	0.1mg/kg	0.2mg/kg
Onset	60 sec	2-3 min	2-3min
Duration	35min	30-35min	60-90 min
Features			Organ-independent metabolism

5. Lisa likely had a rapid sequence induction using suxamethonium as the urgency of the clinical situation dictated that her surgery could not be delayed 6 hours of fasting time. She may also have been nauseated or vomiting prior to induction. Suxamethonium use is associated with post-operative myalgia, particularly in young females. Other contributors to post-operative nausea and vomiting (PONV) could have included the use of volatile agent, N2O, opioids, tramadol, post-operative ileus, sepsis and dehydration.

CASE FOUR

Short case number: 4_01_04

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Hospital Ward

Topic: Introduction to Airway Management

CASE



Judy is a 60 year old female who presents for a right hemicolectomy on tomorrow's list.

A routine colonoscopy found a large caecal polypoid mass. CT chest and abdomen have not shown evidence of any metastases.

Her medical history includes mild asthma and elevated body mass index (BMI). Her height is 170cm, her weight 87kg, and she is of typical gynaecoid ("pear-shaped") appearance. She uses a salbutamol inhaler prn.

QUESTIONS

1. What anaesthetic issues are there to consider arising from this scenario?
2. Outline the components of the pre-operative assessment that will help you manage these issues.
3. Why is Airway assessment important? How will you do this?
4. How do you manage patients with asthma and other reactive airways disease perioperatively?
5. Outline the anatomy of the airway from the mouth to the trachea, including the sensory innervation of the key areas.
6. How do you prepare for airway management in an elective general anaesthetic case?
7. What equipment is available to assist with airway management? Outline how you would manage an airway using basic equipment.
8. What are the indications for endotracheal intubation? What other options exist for airway management during operative procedures.
9. Outline the procedure of endotracheal intubation. Include an outline of the complications associated with endotracheal intubation.

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Duke J 2000 Anaesthesia Secrets 2nd Ed Hanley and Belfus, Inc. Philadelphia
- Morgan et al 2002 Clinical Anaesthesiology 3rd Ed McGraw-Hill New York

Electronic resources [also available on Blackboard]

- PS 18 Recommendations of Monitoring During Anaesthesia
<http://www.anzca.edu.au/resources/professional-documents/>

ANSWERS

What anaesthetic issues are there to consider arising from this scenario?

- optimization of asthma
- elevated BMI
- procedure requires PPV and will be of approx 3hours duration
- post-operative analgesia in presence of laparotomy wound and bowel rehabilitation

Outline the components of the pre-operative assessment that will help you manage these issues?

In addition to the usual information (detailed in week1.1):

- (i) Exclude any clinical manifestation of the presenting complaint such as bowel obstruction or severe anaemia
(usually an asymptomatic presentation)
- (ii) **Asthma: assess severity of disease and treatment history (see above)**
- (iii) **Obesity:** relevant to risk of difficult airway management, ease of venous access, monitoring, patient positioning and pressure area care.

Airway assessment. Why is Airway assessment important?

See case 4.1.1

Airway assessment is critical prior to rendering a patient unconscious and apnoeic .

How do you manage patients with asthma and other reactive airways disease perioperatively?

- (i) **Assessment of severity of disease, recent stability and current status**
ie. when diagnosed, symptoms and triggers, frequency of episodes, last episode, “ Are you as good as you can be?”

- (ii) Treatment history
ie. regular treatment and compliance, use of steroids, last steroid therapy, hospitalization/ICU admission/ventilation, peak flow monitoring

Regular or recent oral steroid therapy may necessitate steroid supplementation in the perioperative period.

- (iii) Management goals
 - optimize pre-op : may include increasing usual therapy or adding pulse oral steroid therapy
 - continue regular therapy
 - prevent triggers for exacerbation
such as unnecessary airway manipulation and exposure to airway irritants (some volatile agents)
 - aggressive post-operative chest physiotherapy and mobilization

Liaise with Respiratory Physician as indicated.
Applies to COAD with reversible component

Outline the anatomy of the airway from the mouth to the trachea, including the sensory innervation of the key areas.

The airway begins with the entry points to the oral and nasal cavities and extends to the trachea. Beyond these cavities, it is divided into the oropharynx, the nasopharynx and the hypopharynx, before becoming the larynx and trachea.

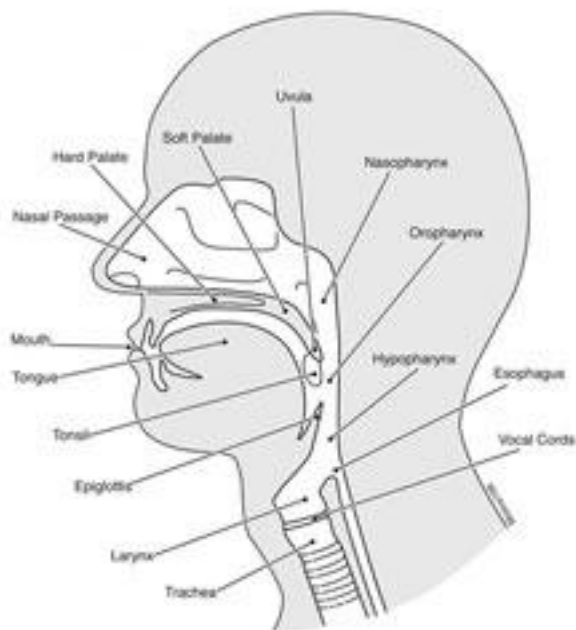
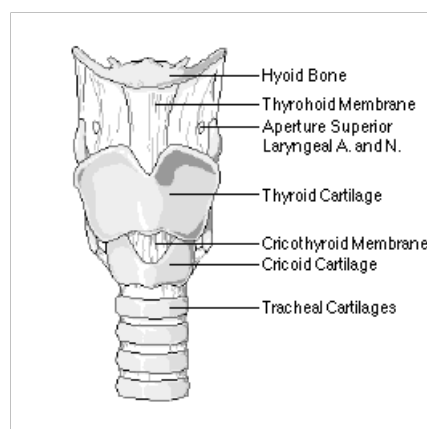


Figure 3 - NORMAL ANATOMY OF UPPER AIRWAY

Anatomy of the Larynx

The larynx is a cylindrical structure composed of 3 unpaired (thyroid, cricoid and epiglottis) and 3 paired cartilages (arytenoid, corniculate and cuneiform). These are held together by ligaments and muscle. The functions of the larynx are to protect the entry to the trachea and enable phonation.



Innervation

Upper airway

CN V1 - anterior ethmoidal nerve to anterior nasal mucous membranes

- CN V2 - sphenopalatine and palatine nerves to posterior nasal mucous membranes, hard and soft palate
- CN V3 - lingual nerve to anterior 2/3 of tongue
- Lower Airway**
- CN IX - glossopharyngeal nerve to the soft palate, pharynx, tonsils, vallecula
- CN X - superior laryngeal nerve to epiglottis and the vocal cords
- recurrent laryngeal nerve to larynx below the vocal cords and trachea

How do you prepare for airway management in an elective general anaesthetic case?

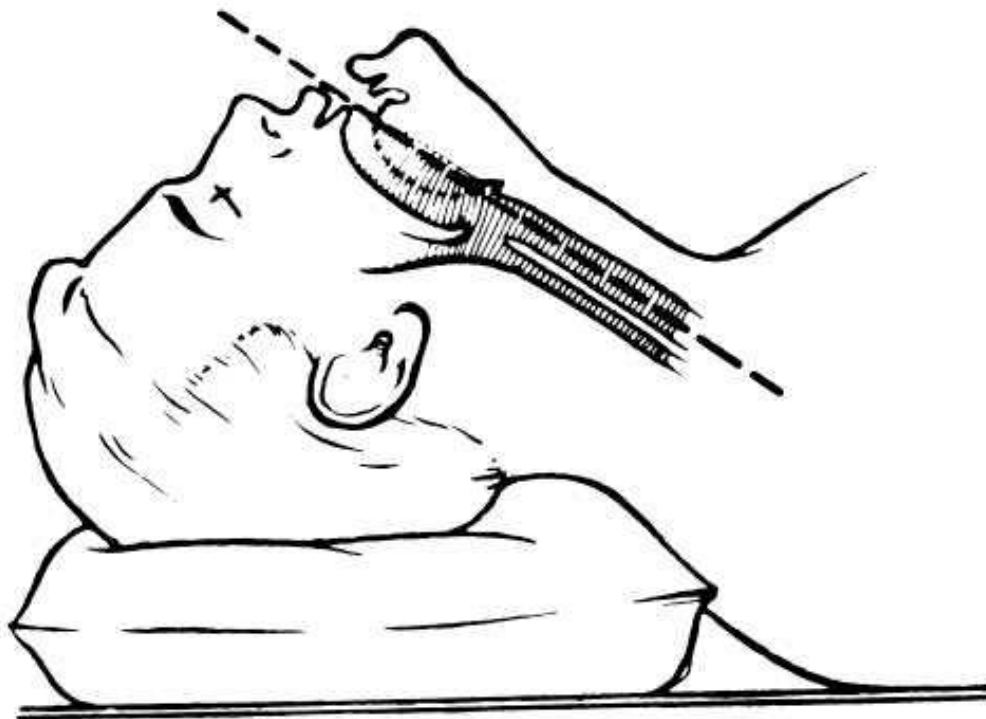
5. Preparation for Airway Management

Airway Management is necessarily the anaesthetist's field of expertise. Following administration of drugs that render a patient apnoeic and unable to protect his/her own airway, it is critical to provide him/her with oxygenation and airway protection.

It is critical to remember that providing **oxygenation** to the patient is the aim, and the means by which this is achieved should always be of secondary concern. (It is very easy to become distracted by the goal of intubation when bag and mask ventilation will allow oxygenation.)

Elective airway management for general anaesthesia must take place under ideal conditions including

- appropriately trained assistant present
- adjustable trolley
- **standard monitoring equipment: SpO2, NIBP, ECG, oxygen and CO2 analyser, breathing system disconnect/ventilation failure alarm, anti-hypoxia device**
- **suction**
- access to equipment as listed below
- **established venous access** (with few exceptions)
- **optimal head position "sniffing the morning air"**
- preoxygenation of patient



Essential to preparation is a clear plan that is communicated to all involved. All team members should take universal precautions for the prevention of disease transmission.

What equipment is available to assist with airway management? Outline how you would manage an airway using basic equipment.

Oxygen/Gas delivery system

Cuff syringe

Face masks

Devices to maintain a patent airway without endotracheal intubation : oral and nasopharyngeal airways, laryngeal mask airways (LMA)

Endotracheal tubes (ETT)

Laryngoscopes and blades

Accessory devices: bougie, stylet and Magill forceps

Advanced airway devices: video laryngoscopes, fiberoptic bronchoscopes and other supraglottic airways

Cricothyroidotomy kit

Students should familiarize themselves with the appearance and use of this equipment in the operating theatres.

Key points:

Standard Adult ETT sizes: male 8.0, female 7.0

Standard adult LMA sizes: male 4-5, female 3-4

Standard adult laryngoscope blade: Macintosh 3

Nasal airway/ intubation is contraindicated in patients with a bleeding diathesis

What are the indications for endotracheal intubation? What other options exist for airway management during operative procedures.

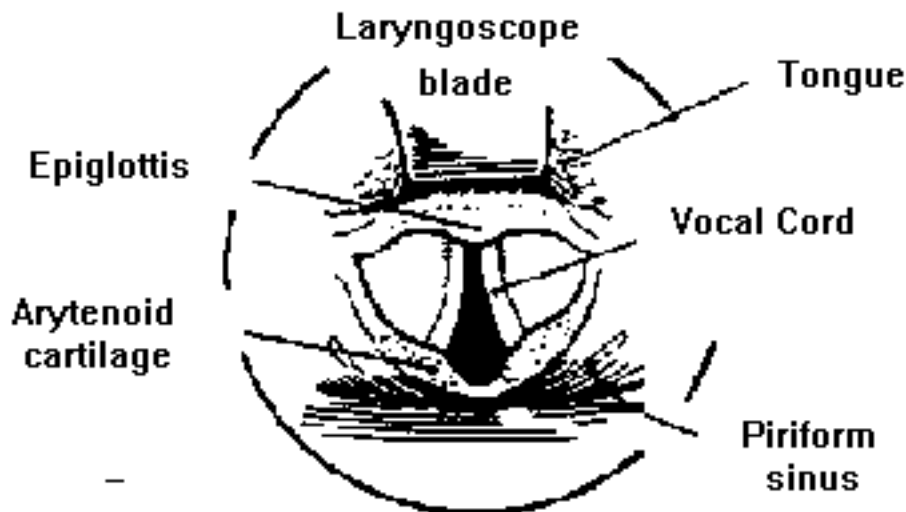
Indications

- protection of the respiratory tract from aspiration
- requirement for PPV
- long duration of surgery
- secure the airway in various surgical positions, or when surgery is in close proximity to the airway
- control PaCO₂
- require tracheobronchial toilet
- lung isolation
- respiratory failure

Outline the procedure of endotracheal intubation. Include an outline of the complications associated with endotracheal intubation.

Following intravenous induction, unconsciousness and apnoea is confirmed in the patient before beginning face mask ventilation. This will continue until enough time has elapsed for the muscle relaxant to take effect and provide optimal intubating conditions. During this time, the patient is ventilated with 100% O₂ and volatile agent to maintain anaesthesia.

The laryngoscope is taken with the left hand, passed down the right hand side of the patient's oral cavity to sweep the tongue to the left. Simultaneously, pressure is applied toward the patient's toes to expose the larynx. The view at laryngoscopy is illustrated below.



This is the optimum view, however lesser views are illustrated below in the Cormack and Lehane grading system. This view, and any other difficulties with airway management, should be documented for the benefit of subsequent anaesthetists on the anaesthetic record.

Cormack and Lehane Laryngoscopic View Grades (b)

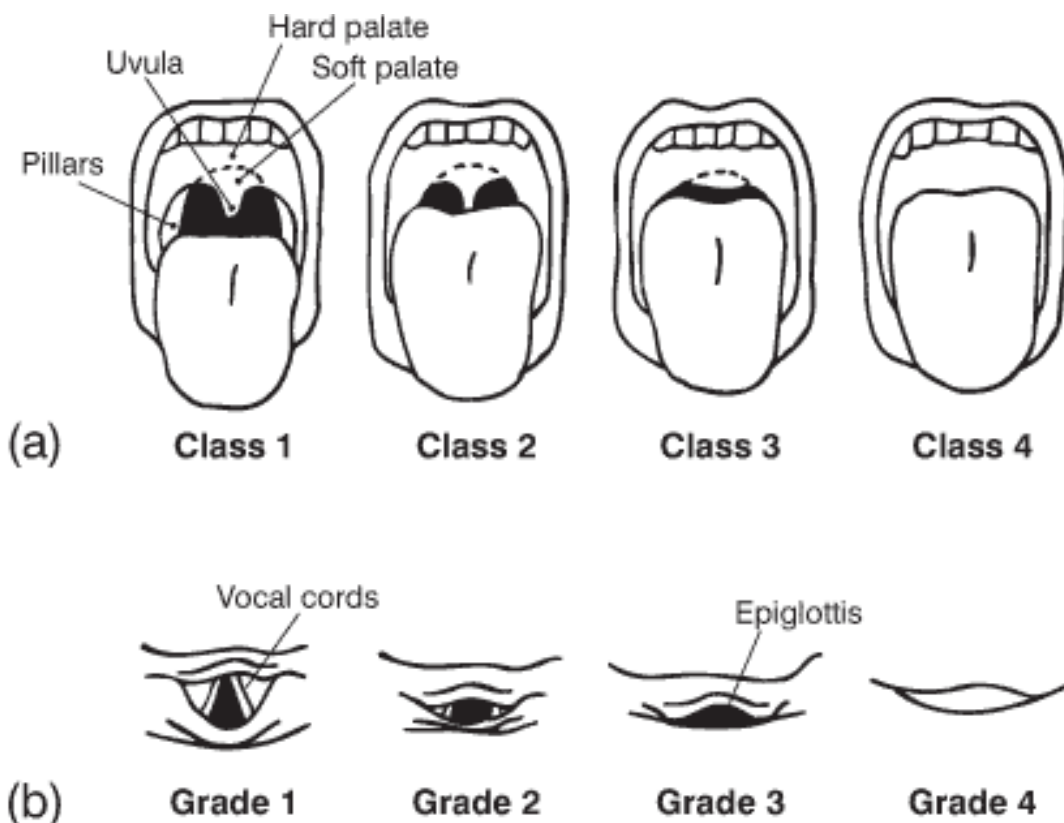


Figure 2 - a) Mallampati classification modified by Samsoon and Young: Class 1 - visualization of the soft palate, Class 2 - complete visualization of uvula, Class 3 - visualization of the base of the uvula, Class 4 - soft palate is not visible at all; **b)** Laryngoscopy according to the classification of Cormack and Lehane: Grade I - most of the glottis visible, Grade II - only the posterior extremity of the glottis visible, Grade IV - not even the epiglottis visible⁶

Once the optimum view is obtained, the ETT is taken with the right hand and passed between the vocal cords up to the indicator line on the ETT. The cuff is then inflated and the circuit connected to the ETT. Cuff volume should be the minimum volume that prevents audible leak at ventilation pressures between 25 to 30cm H₂O for adults.

Confirmation of correct ETT placement

- visualise ETT pass through the vocal cords
- observe chest rise and fall with bag ventilation
- confirm appropriate end-tidal CO₂ wave-form
- confirm bilateral and equal breaths sounds on chest auscultation
- check peak inspiratory pressures

Note length of ETT at teeth and secure ETT.

Potential Complications of Endotracheal Intubation

Malpositioning: oesophageal or endobronchial intubation, unintentional extubation

Airway Trauma: dental damage, lacerations, sore throat, vocal cord injury/granuloma, nerve palsies

Physiologic Responses: hypertension, tachycardia, laryngospasm, raised intracranial and intraocular pressure

ETT malfunction: cuff tear or herniation

Cuff injury: tracheal mucosal ulceration/ischaemia

CASE FIVE

Short case number: 4_01_05

Category: Clinical Skills

Discipline: Anaesthesia

Setting: Emergency Department

Topic: Emergency Anaesthesia_airway management.

CASE

The orthopaedic registrar calls to book a 25 year old male for an Open Reduction Internal Fixation (ORIF) of a compound fracture of right distal tibia and fibula sustained in a motorbike accident (MBA). He is concerned that the vascular supply is compromised as distal pulses are weak. The patient also has facial lacerations including to his upper lip that require a compound scrub and repair. Both leg and facial wounds are heavily soiled and the surgeon would like to proceed as soon as possible.

The patient had been drinking alcohol prior to the accident and had a blood alcohol content of 0.07 on presentation 1 hour ago. He is not known to have any past medical history, states no known allergies and is a current smoker.

The orthopaedic registrar has since cleared his cervical spine of injury and removed the hard collar.

He has had a total of 20mg of morphine over the past 1.5 hours.

QUESTIONS

1. Describe your approach to preparing this patient for a general anaesthetic.
2. What features of the case contribute to his aspiration risk?
3. You suspect this will be a difficult airway to manage. What features on history and examination may indicate that this is a "difficult airway"?
4. How will you prepare?
5. What are the principles of a rapid sequence induction?
6. Outline an appropriate plan in the event of a failed intubation. What are the principles of failed intubation planning?

Resources:

- Aitkenhead, AR, Smith, G, & Rowbotham, DJ (2007). Textbook of Anaesthesia (5th ed.). London: Churchill Livingstone.
- Padley, AP (2008). Westmead pocket anaesthetic manual (3rd ed.) Sydney: McGraw Hill.

Online Resources:

Difficult Airway Society: <http://www.das.uk.com/guidelines/downloads.html>

ANSWERS

Describe your approach to preparing this patient for a general anaesthetic.

Owing to the nature of this patient's injuries, his surgery cannot be delayed to allow appropriate fasting (and sobering up) time to elapse. Prompt exploration of the wound may restore the vascular supply and prevent ischaemic damage to the distal tissue. Expediting debridement of soiled and devitalized tissues is key to minimizing infection risk.

The anaesthetist should perform the usual pre-anaesthetic assessment including history, examination and review of investigations. In urgent trauma cases, the Advanced Trauma Life Support (ATLS) system provides a comprehensive framework for simultaneous diagnosis and management. The anaesthetist should review the workup undertaken so far by the Emergency Medicine team.

Primary Survey as per ATLS Guidelines

Airway maintenance with cervical spine protection

Breathing and ventilation

Circulation with haemorrhage control

Disability: Neurologic status

Exposure/Environmental Control: completely undress patient but prevent hypothermia

Secondary Survey

History:

Allergies

Medications currently used

Past illness/Pregnancy

Last meal

Events/Environment related to injury

Physical Examination: head to toe examination including logroll, perineal region PRE +/- PVE
"fingers and tubes in every orifice"

Investigations: ECG and trauma series of x-rays + further imaging, ABG's, pregnancy test, toxicology screen as indicated

Additional information relevant to the anaesthetist:

History of presenting complaint:

(1) Nature of MBA: speed, combatant (eg. car, tree), point of impact (eg. thrown onto road), protective clothing (eg. presence of helmet, leathers), state on scene and in transit

(2) Alcohol and other intoxicant intake including quantity and time frame.

(3) Background medical, surgical and anaesthetic history, smoking, alcohol use, medication, allergies and fasting status.

Examination

Special attention to:

(1)General status: distressed, intoxicated, altered conscious state, co-operative, hydration, observations, ability to consent

(2)Airway assessment:

Perioral and other facial lacerations hint at

- cervical spinal trauma
- intra-oral trauma including broken teeth, tongue or mucosal lacerations which add to airway difficulties (unstable cervical spine, aspiration of tooth fragments, stomach full of swallowed blood, swollen tongue)

(3)Potential occult injuries

What features of the case contribute to his aspiration risk?

- alcohol intoxication
- not fasted
- trauma, pain
- blood in airway
- recent opioid administration

You suspect this will be a difficult airway to manage. What features on history and examination may indicate that this is a “difficult airway”?

His airway management will be difficult because of

- high aspiration risk
- perioral lacerations may limit mouth opening or impede mask seal
- potential blood in airway, intraoral lacerations or swelling

The patient’s background risk of airway difficulty must be taken into account with the usual assessment of indicators such as Mallampatti score etc.

To optimize the patient’s preparation:

- explain the procedure and allay anxiety
- prokinetic agent to empty stomach
- ensure optimal positioning

Equipment required includes an anaesthetic delivery system, suction and monitoring, an appropriately sized face mask, size 3 and 4 Mackintosh laryngoscope blades with a light source handle, a variety of sizes of cuffed endotracheal tubes, a stylet and bougie and 10 ml cuff syringe. The difficult intubation trolley should be in the room to allow access to airway devices, such as Guedel airways, LMA’s, bronchoscopes and alternate laryngoscope blades. All equipment must be checked for function and within easy reach as once your assistant applies cricoid pressure, he/she cannot leave to fetch other equipment.

In the case of a suspected difficult airway, the plan for management should be clearly communicated to the operating theatre team. Your assistant should be clear on your initial plan (Plan A) for management and the fallback positions if this fails (Plans B and C etc).

What are the principles of a rapid sequence induction?

This patient will require a rapid sequence induction for endotracheal intubation.

The principles of this are

- preoxygenation

- optimal patient position
- application of cricoid pressure on induction
- rapid intubation to secure the airway
- avoidance of bag-mask ventilation
- use of short-acting agents that allow return of spontaneous ventilation in an acceptable time frame if ventilation and airway control is not possible

If the larynx is unable to be intubated despite appropriate manoeuvres, bag and mask ventilation should be instituted and a decision taken whether to awaken the patient or maintain anaesthesia. Many algorithms exist to assist with this decision making process, including this from the Difficult Airway Society.

Outline an appropriate plan in the event of a failed intubation. What are the principles of failed intubation planning?

